

STAT 5309

LAB 4-A

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A. 2-FACTOR DESIGN

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Due: Thurs, Feb 21

A. 2-FACTOR DESIGN

We consider 2 separate factors, with some interaction between the two.

Data: concrete. (Compaction Effects on Asphalt Concrete Durability): Asphalt pavements suffer from water-associated deteriorations such as crackings. Research to find better pavements more resistant to deterioration. 2 factors to have an effect : (1) compact methods used to compact the specimen (2) aggregate type used in the asphalt mixture. There are 4 compact methods and 2 aggregate types,

Type	Kneading (4 levels)				
	Static	Regular	Low	Very low	
Basalt	68	126	93	56	
	63	128	101	59	
	65	133	98	57	
Silicious	71	107	63	40	
	66	110	60	41	
	66	116	59	44	

“Type” has 2 levels. “Kneading” has 4 levels. Total;u, there are 8 combinations. Each combination has 3 replicates. Totally, 24 observations.

```
#-----Set up data frame-----  
type <- rep(c(1,2),each=12)  
meth <- rep(c(1,2,3,4),each=3,times=2)  
strength <- c(68,63,65,126,128,133,93,101,98,56,59,57,  
              71,66,66,107,110,116,63,60,59,40,41,44)  
  
concrete <- data.frame(type,meth,strength)  
attach(concrete)
```

```
type <- factor(type,levels=1:2,labels=c("B","S"))
meth <- factor(meth,levels=1:4,labels=c("S","R","L","V"))
```

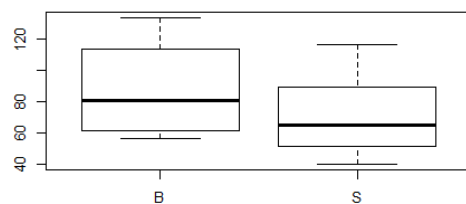
```
tapply(strength,list( type,meth),mean)
```

	S	R	L	V
B	65.33333	129	97.33333	57.33333
S	67.66667	111	60.66667	41.66667

```
tapply(strength,list( type,meth),sd)
```

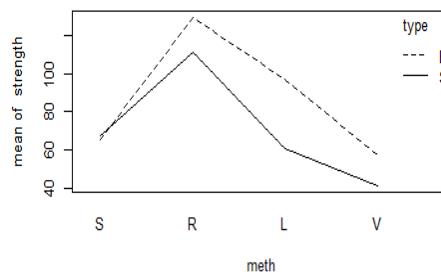
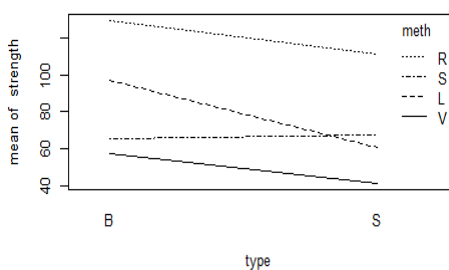
	S	R	L	V
B	2.516611	3.605551	4.041452	1.527525
S	2.886751	4.582576	2.081666	2.081666

```
#-----Box plot; Interaction plot-----
boxplot(strength ~ type)
```



```
interaction.plot(type,meth, strength)
Interaction.plot(meth, type, strength)
```

Note: There are **interactions**



```
#----- Model with Interaction-----
```

```
concrete.mod <- aov(strength ~ type* meth) # interaction term
summary.lm(concrete.mod)
```

	Df	Sum Sq	Mean Sq	F value	Pr(>F)
type	1	1734	1734	182.53	3.63e-10 ***
meth	3	16243	5414	569.95	< 2e-16 ***
type:meth	3	1145	382	40.17	1.12e-07 ***
Residuals	16	152	10		

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

anova(concrete.mod)

#ANOVA analysis, which is the same as above

Analysis of Variance Table

Response: strength

	Df	Sum Sq	Mean Sq	F value	Pr(>F)
type	1	1734	1734.0	182.526	3.628e-10 ***
meth	3	16244	5414.5	569.947	< 2.2e-16 ***
type:meth	3	1145	381.7	40.175	1.124e-07 ***
Residuals	16	152	9.5		

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Notes:

- (1) Factor levels for type, meth and type-meth are significant different.
- (2) On degrees of freedom.

$n(\text{number of replicates}) = 3, \text{levels: } \text{type}(a = 2); \text{meth}(b = 4). \text{Total: } N = a * b * n = 24$

$Df(\text{type}) = a - 1 = 2 - 1 = 1$;

$Df(\text{meth}) = b - 1 = 4 - 1 = 3$

$Df(\text{type:meth}) = (a - 1)(b - 1) = 3$

$Df(\text{Residuals}) = df(SST) - Df(\text{type}) - Df(\text{meth}) - Df(\text{type:meth})$

$= abn - (a - 1) - (b - 1) - (a - 1)(b - 1) = abn - ab = ab(n - 1)$

$= 2 * 4 * (3 - 1) = 16$

##-----Multiple Comparison: TukeyHSD-----

Tukey <- TukeyHSD(concrete.mod)

```
> Tukey
Tukey multiple comparisons of means
 95% family-wise confidence level

Fit: aov(formula = strength ~ type * meth)

$type
      diff      lwr      upr p adj
S-B  -17 -19.66749 -14.33251    0

$meth
      diff      lwr      upr      p adj
```

```
R-S 53.5 48.408778 58.59122 0.00e+00
L-S 12.5 7.408778 17.59122 1.56e-05
V-S -17.0 -22.091222 -11.90878 3.00e-07
L-R -41.0 -46.091222 -35.90878 0.00e+00
V-R -70.5 -75.591222 -65.40878 0.00e+00
V-L -29.5 -34.591222 -24.40878 0.00e+00
```

plot(Tukey)

##-----Multiple comparison; LSD.test(), Package “agricolae”

```
> LSD.test(concrete.mod, c("meth", "type"), console=T)
```

```
Study: concrete.mod ~ c("meth", "type")
```

```
LSD t Test for strength
```

```
Mean Square Error: 9.5
```

```
meth:type, means and individual ( 95 %) CI
```

	strength	std r	LCL	UCL	Min	Max
L:B	97.33333	4.041452 3	93.56093	101.10573	93	101
L:S	60.66667	2.081666 3	56.89427	64.43907	59	63
R:B	129.00000	3.605551 3	125.22760	132.77240	126	133
R:S	111.00000	4.582576 3	107.22760	114.77240	107	116
S:B	65.33333	2.516611 3	61.56093	69.10573	63	68
S:S	67.66667	2.886751 3	63.89427	71.43907	66	71
V:B	57.33333	1.527525 3	53.56093	61.10573	56	59
V:S	41.66667	2.081666 3	37.89427	45.43907	40	44

```
Alpha: 0.05 ; DF Error: 16
```

```
Critical Value of t: 2.119905
```

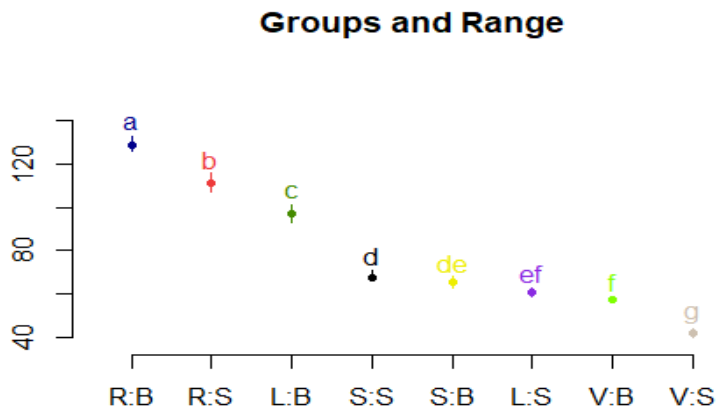
```
least Significant Difference: 5.334978
```

```
Treatments with the same letter are not significantly different.
```

	strength groups
R:B 129.00000	a
R:S 111.00000	b
L:B 97.33333	c
S:S 67.66667	d
S:B 65.33333	de
L:S 60.66667	ef
V:B 57.33333	f
V:S 41.66667	g

```
LSD <- LSD.test(concrete.mod, c("meth", "type"), console=T)
```

```
Plot(LSD)
```



```
##-----Power- Sample size-----
#-----Fpower2(): Power of the F-test for 2 factors-----

# MSE=9.5 , sigma= 3.08
> sqrt(9.5)
[1] 3.082207

rmin <- 2
rmax<- 10
alpha<- .05
sigma <- 3.08
Delta <- 12.5 # minimum marginal means difference of meth needed to be detected, use Tukey
nlev <- c(2,4)
nreps<-c(rmin, rmax)
power <-Fpower2(alpha, nlev, nreps, Delta, sigma)
power
```

	alpha	a	b	nreps	Delta	sigma	powera	powerb
[1,]	0.05	2	4	2	12	3.16	0.9999975	0.9999997
[2,]	0.05	2	4	10	12	3.16	1.0000000	1.0000000

Note: n=3 replicates is a satisfactory number.

```
## -----Contrasts/ -----
##-----estimable()-----

library(gmodels)
cont<- rbind(
  c(1,-1,0,0,0,0,0,0), c(0,0,1,-1,0,0,0,0), c(0,0,0,0,1,-1,0,0), c(0,0,0,0,0,0,1,-1))
```

```
rownames(cont) <- c("B vs S in Static", "B vs S in Regular", "B vs S in Low", "B vs S in Very Low")
```

```
> estimable(concrete.mod,cont, conf.int=.95)
```

	Estimate	Std. Error	t value	DF	Pr(> t)	Lower.CI	Upper.CI
B vs S in Static	-2.333333	2.516611	-0.9271726	16	3.676127e-01	-7.668311	3.001645
B vs S in Regular	18.000000	2.516611	7.1524747	16	2.296669e-06	12.665022	23.334978
B vs S in Low	36.666667	2.516611	14.5698559	16	1.181086e-10	31.331689	42.001645
B vs S in Very Low	15.666667	2.516611	6.2253021	16	1.214107e-05	10.331689	21.001645

Note: B and S are significant in all , **except in Statics** (CI interval contains 0)

```
> 67.67 - 65.33
```

```
[1] 2.34
```

B. ONE BLOCKING FACTOR

```
# ----- Randomized Complete Block Design (RCBD) - -----
```

```
# Create a RCBD design: : design.rcbd(); package "agricolae" -----
```

Data: Crops of flowers. Grouped into kinds.

```
library(agricolae)
```

```
treat<- c(1,2,3,4) #treatments
```

```
design.rcbd <- design.rcbd(treat, r=4, seed=11)
```

```
book.rcbd <- design.rcbd$book
```

```
levels(book.rcbd$block) <- c("carnation", "daisy", "rose", "tulip") #change names of blocks
```

```
book.rcbd
```

	plots	block	treat
1	101	carnation	4
2	102	carnation	1
3	103	carnation	2
4	104	carnation	3
5	201	daisy	2
6	202	daisy	4
7	203	daisy	3
8	204	daisy	1
9	301	rose	4
10	302	rose	2
11	303	rose	1
12	304	rose	3
13	401	tulip	1
14	402	tulip	4
15	403	tulip	2
16	404	tulip	3

#Linear model with RCBD-----

Data: drug. Rat Behavior. 50 observations.

Rat: a factor with levels 1 2 3 4 5 6 7 8 9 10. [There are 10 rats]

Dose: a factor with levels 0.0, 0.5, 1.0, 1.5, 2.0.

Rate: a numeric vector

```
library(daewr)
```

```
data(drug)
```

```
head(drug)      factors : rat and dose; rat serves as a blocking factor
```

```
> drug
  rat dose rate
1   1    0 0.60
2   1   0.5 0.80
3   1    1 0.82
4   1   1.5 0.81
13  3    1 0.83
14  3   1.5 0.80
15  3    2 0.52
16  4    0 0.60
47 10   0.5 1.20
48 10    1 1.18
49 10   1.5 1.23
50 10    2 1.05
```

```
>attach(drug)
```

```
> levels(rat)      #10 levels
```

```
[1] "1" "2" "3" "4" "5" "6" "7" "8" "9" "10"
```

```
> levels(dose)     #5 levels
```

```
[1] "0" "0.5" "1" "1.5" "2"
```

```
drug.mod1<- aov(rate ~ rat +dose, data=drug) #Suppose no interaction between rat and dose
```

```
summary.aov(drug.mod1)
```

	Df	Sum Sq	Mean Sq	F value	Pr(>F)
rat	9	1.6685	0.18538	22.20	3.75e-12 ***
dose	4	0.4602	0.11505	13.78	6.53e-07 ***
Residuals	36	0.3006	0.00835		

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Note: Effectiveness of blocking factor:

$MSB/MSE = 0.18538 / \underline{0.00835} = 22.20 \gg 1$. Blocking is effective

```
summary.lm(drug.mod1)
```

```
> summary.lm(drug.mod1)
```

```
Call:
```

```
aov(formula = rate ~ rat + dose, data = drug)
```

```

Residuals:
    Min       1Q   Median       3Q      Max
-0.18780 -0.04155 -0.00380  0.05670  0.15620

Coefficients:
            Estimate Std. Error t value Pr(>|t|)
(Intercept)  0.914200   0.012922  70.749 < 2e-16 ***
rat1         -0.208200   0.038765  -5.371 4.82e-06 ***
rat2         -0.222200   0.038765  -5.732 1.59e-06 ***
rat3         -0.196200   0.038765  -5.061 1.25e-05 ***
rat4         -0.092200   0.038765  -2.378  0.0228 *
rat5          0.073800   0.038765   1.904  0.0650 .
rat6         -0.080200   0.038765  -2.069  0.0458 *
rat7         -0.006200   0.038765  -0.160  0.8738
rat8          0.231800   0.038765   5.980 7.40e-07 ***
rat9          0.291800   0.038765   7.527 6.76e-09 ***
dose.L        0.078108   0.028894   2.703  0.0104 *
dose.Q       -0.198575   0.028894  -6.873 4.83e-08 ***
dose.C       -0.020239   0.028894  -0.700  0.4882
dose^4       -0.008845   0.028894  -0.306  0.7613
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 0.09137 on 36 degrees of freedom
Multiple R-squared:  0.8763,    Adjusted R-squared:  0.8316
F-statistic: 19.61 on 13 and 36 DF, p-value: 1.59e-12

```

##-----Without using Rat as blocking factor-----

```

drug.mod2 <- aov(rate ~ dose)
summary(drug.mod2)

```

	Df	Sum Sq	Mean Sq	F value	Pr(>F)
dose	4	0.4602	0.11505	2.629	0.0466 *
Residuals	45	1.9690	0.04376		

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Note: $MSE(drug.mod2) = 0.04376$, compare $MSE(drug.mod1) = 0.00835$
So 1-Blocking factor reduces the MSE significantly.(80%)
 $> (0.00835 - 0.04376) / 0.04376$
[1]

0.8091865

C. EXERCISES

1.

A chemist wishes to test the effect of four chemical agents on the strength of a particular type of cloth. Because there might be variability from one bolt to another, the chemist decides to use a randomized block design, with the bolts of cloth considered as blocks. She selects five bolts and applies all four chemicals in random order to each bolt. The resulting tensile strengths follow. Analyze the data from this experiment (use $\alpha = 0.05$) and draw appropriate conclusions.

Chemical	Bolt				
	1	2	3	4	5
1	73	68	74	71	67
2	73	67	75	72	70
3	75	68	78	73	68
4	73	71	75	75	69

- (a) Create a vector for Blocks, named "Bolt": 5 levels. Total 20.
Create a vector for Treatments, named "Chemical". Total 20.
Create a response vector, named "Strength". Set up the data frame, named "chem".
- (b) Any evidence that the Chemical affect Strength ?
- (c) Perform a TukeyHSD to compare the treatment means. Which Chemical is the preferred(bring the highest strength)
- (d) Check the assumption of the residuals.

2.

Three different washing solutions are being compared to study their effectiveness in retarding bacteria growth in 5-gallon milk containers. The analysis is done in a laboratory, and only three trials can be run on any day. Because days could represent a potential source of variability, the experimenter decides to use a randomized block design. Observations are taken for four days, and the data are shown here. Analyze the data from this experiment (use $\alpha = 0.05$) and draw conclusions.

Solution	Days			
	1	2	3	4
1	13	22	18	39
2	16	24	17	44
3	5	4	1	22

- (e) Create a vector for Blocks, named "Days": 4 levels. Total 12.
Create a vector for Treatments, named "Solutions". Total 12
Create a response vector, named "Growth". Set up the data frame.
- (f) Any evidence that the Solutions affect bacteria growth ?
- (g) Perform a TukeyHSD to compare the treatment means. Which Solution is preferred (brings the lowest bacterial growth)
- (h) Check the assumption of the residuals.

3.

An aluminum master alloy manufacturer produces grain refiners in ingot form. The company produces the product in four furnaces. Each furnace is known to have its own unique operating characteristics, so any experiment run in the foundry that involves more than one furnace will consider furnaces as a nuisance variable. The process engineers suspect that stirring rate affects the grain size of the product. Each furnace can be run at four

different stirring rates. A randomized block design is run for a particular refiner and the resulting grain size data is shown below:

Stirring Rate (rpm)	Furnace			
	1	2	3	4
5	8	4	5	6
10	14	5	6	9
15	14	6	9	2
20	17	9	3	6

- (a) Is there any evidence that stirring rate affects grain size?
- (b) Graph the residuals from this experiment on a normal probability plot. Interpret this plot.
- (c) Plot the residuals versus furnace and stirring rate. Does this plot convey any useful information?
- (d) What should the process engineers recommend concerning the choice of stirring rate and furnace for this particular grain refiner if small grain size is desirable?